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Optimal Estimation Theory for Dynamic Systems with Set Membership Uncertainty: An Overview*

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A review of the main results of set membership estimation theory shows that theoretical results and algorithms are now available, and can be applied to practical problems where a statistical approach is questionable.

Key Words—Estimation theory; deterministic analysis; error analysis; identification; minmax techniques; parameter estimation; performance bounds; robustness; set membership uncertainty; unknown but bounded errors.

Abstract—In many problems such as linear and nonlinear regressions, parameter and state estimation of dynamic systems, state-space and time series prediction, interpolation, smoothing and functions approximation, one has to evaluate some unknown variable using available data. The data are always associated with some uncertainty and it is necessary to evaluate how this uncertainty affects the estimated variables. Typically, the problem is approached assuming a probabilistic description of uncertainty and applying statistical estimation theory. An interesting alternative approach, referred to as set membership or unknown but bounded (UBB) error description, has been investigated since the late 1960s. In this approach, uncertainty is described by an additive noise which is known only to have given integral (typically l_2 of l_1) or componentwise (l_∞) bounds. In this paper we review the main results of this theory, with special attention to the most recent advances obtained in the case of componentwise bounds.

INTRODUCTION

ESTIMATION THEORY is concerned with the problem of evaluating some unknown variables depending on given data (often obtained by measurements on a real process). Available data are always known with some uncertainty and it is necessary to evaluate how this uncertainty affects the estimated variables.

Obviously the solution of the problem depends on the type of assumptions made on uncertainty. The cases most investigated so far are unquestionably related to the assumption that uncertainty is given by an additive random noise with (partially) known probability density function (pdf). However, in many situations the very random nature of uncertainty may be questionable. For example, the real process generating the actual data may be very complex (large scale, nonlinear, time-varying) so that only simplified models can be practically used in the estimation process. Then, the residuals of the estimated model have a component due to deterministic structural errors, and treating them as purely random variables may lead to unsatisfactory results.

An interesting alternative approach, referred to as set membership or unknown but bounded (UBB) error description, has been pioneered by the work of Witsenhausen and Schweppe in the late 1960s (Witsenhausen, 1968; Schweppe, 1968, 1973). In this approach, uncertainty is described by means of an additive noise which is known only to have given bounds. Motivation for this approach is the fact that in many practical cases the UBB error description is more realistic and less demanding than the statistical description. However, despite the appeal of its features, the UBB approach has not reached a wide diffusion yet. An important reason for this is certainly the fact that until the first 1980s reasonable results and algorithms had been obtained only for uncertainty bounds of integral type (mainly l_2), while in practical applications componentwise bounds (l_∞) are mainly of interest.

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Real advances have been obtained in the last few years for the componentwise bounds case, leading to theoretical results and algorithms which can be properly applied to practical problems where the use of statistical techniques is questionable.

The purpose of this paper is to review these results and to present them in a unified framework in order to contribute the present state of the art in the field and to stimulate further basic and applied researches.

PROBLEM FORMULATION

In this section we formulate a general framework such that the main results in the literature can be presented in a unifying view. Such formulation can be sketched as follows (Traub and Wozniakowski, 1980; Milanese and Tempo, 1985).

We have a problem element λ (for example a dynamic system or a time function) and we are interested in evaluating some function $S(\lambda)$ of this problem element (for example some parameter of the dynamic system or particular value of the time function). We suppose to know λ not exactly, but have only some information on it. In particular we assume that it is an element of a set K of possible problem elements and that some function $F(\lambda)$ is measured. Moreover, we suppose that exact measurements are not available and actual measurements y are corrupted by some error ρ.

The estimation problem consists in finding an estimator ϕ providing an approximation $\phi(y) \approx S(\lambda)$ using the available data y and evaluating some measure of such approximation. A geometric sketch is shown in Fig. 1.

To be more precise, let us introduce some definitions.

Spaces and operators

Let Λ be a linear normed n-dimensional space over the real field. Consider a given operator S, called a solution operator, which maps Λ into Z

$$S:\Lambda \to Z \tag{1}$$

where Z is a linear normed l-dimensional space over the real field. Our aim is to estimate an element $S(\lambda)$ of the space Z, knowing approximate information about the element λ. We suppose that two kinds of information may be available. The first one (often referred to as *a priori* information) is expressed by assuming that $\lambda \in K$, where K is a subset of Λ. In most cases K is given as:

$$K = \{\lambda \in \Lambda;\ \|R(\lambda - \lambda_0)\| \le 1\} \tag{2}$$

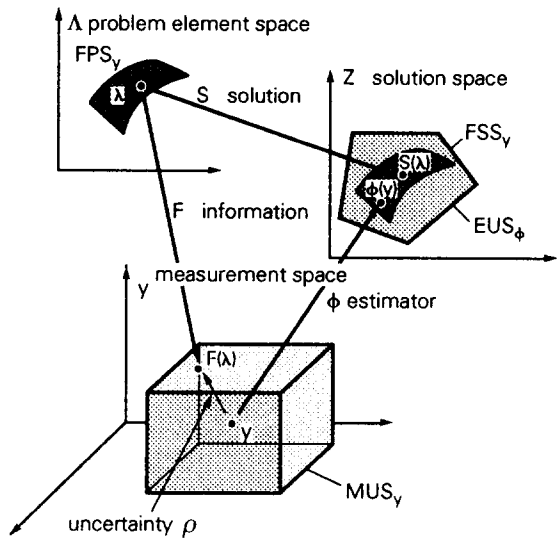

FIG. 1. Generalized estimation problem. MUS_y = Measurement Uncertainty Set, EUS_Φ = Estimate Uncertainty Set, FPS_y = Feasible Problem elements Set and FSS_y = Feasible Solutions Set.

where R is a linear operator and λ_0 is a known problem element. The second kind of information is usually provided by the knowledge of some function $F(\lambda)$, where F, called information operator, maps Λ into a linear normed m-dimensional space Y

$$F:\Lambda \to Y. \tag{3}$$

Spaces Λ, Z, Y are called problem element, solution and measurement spaces, respectively. In the following, unless otherwise specified, we assume that Λ and Z are equipped with l_∞ norms and Y is equipped with an l_∞^w norm.*

In general, due to the presence of noise, exact information $F(\lambda)$ about λ is not available and only perturbed information y is given. In this context, uncertainty is assumed to be additive, i.e.

$$y = F(\lambda) + \rho \tag{4}$$

where the error term ρ is unknown, but bounded by some given positive number ϵ

$$\|\rho\| \le \epsilon. \tag{5}$$

Note that if an l_∞^w norm in measurement space Y is used, componentwise bounds with different values on every measurement can be treated.

An algorithm ϕ is an operator (in general nonlinear) from Y into Z

$$\phi: Y \to Z \tag{6}$$

i.e., it provides an approximation $\phi(y)$ of $S(\lambda)$

* The l_∞^w norm is defined as

$$\|y\|_\infty^w = \max_i w_i\,|y_i|, \quad w_i > 0.$$

using the available data y. Such an algorithm is also called estimator.

Some examples are now presented in order to show how specific estimation problems fit into this general framework.

Example 1. Parameter estimation of ARMA models. Let us consider the ARMA model

$$y_k = \sum_{i=1}^{p} \nu_i y_{k-i} + \sum_{i=1}^{q} \theta_i u_{k-i} + \rho_k \tag{7}$$

where y_k is a scalar output, u_k is a known scalar input and ρ_k is an unknown but bounded error such that

$$|\rho_k| \leq \epsilon_k, \quad \forall k. \tag{8}$$

Suppose that m values $[y_1, \ldots, y_m]$ are known and the aim is to estimate parameters $[\nu, \theta]$. For the sake of simplicity we suppose that $p \geq q$.

Λ can be defined as the $(p+q)$-dimensional space with elements

$$\lambda = [\nu_1, \ldots, \nu_p, \theta_1, \ldots, \theta_q]^T. \tag{9}$$

If no "*a priori*" knowledge on parameter λ is available, then $K = \Lambda$.

Z is the $(p+q)$-dimensional space with elements $z = \lambda$, so that $S(\lambda)$ is identity.

Y is the $(m-p)$ dimensional space with elements $y = [y_{p+1}, \ldots, y_m]^T$, and consequently it results that $F(\lambda)$ is linear and is given by

$$F(\lambda) = \begin{pmatrix} y_p & \cdots & y_1 & u_p & \cdots & u_{p+1-q} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ y_{m-1} & \cdots & y_{m-p} & u_{m-1} & \cdots & u_{m-q} \end{pmatrix} \lambda. \tag{10}$$

Example 2. State estimation of linear dynamic systems. Let us consider the problem of estimating the state of the following discrete, linear, time-invariant dynamic system

$$\begin{cases} x_{k+1} = Ax_k + Bu_k \\ y_k = Cx_k + \rho_k \end{cases} \tag{11}$$

where x_k, y_k, u_k and ρ_k are the state, observation, process noise and observation noise vectors, respectively; A, B and C are given matrices. For the sake of notation simplicity, we suppose that x is l-dimensional and y, u, ρ are scalar variables.

We assume that the samples of process and observation noise are unknown but bounded

$$|u_k| \leq U_k, \quad \forall k \tag{12}$$

$$|\rho_k| \leq \epsilon_k, \quad \forall k. \tag{13}$$

Suppose that m values $[y_1, \ldots, y_m]$ are known and the aim is to estimate x_m. Λ can be defined as the $l + m - 1$-dimensional space with elements

$$\lambda = [x_1^T, u_1, \ldots, u_{m-1}]^T. \tag{14}$$

If no *a priori* information on the initial state x_1 is available, K is defined by

$$K = \{\lambda \in \Lambda; |u_j| \leq U_j, j = 1, \ldots, m-1\}. \tag{15}$$

Z is the l-dimensional space with elements $z = x_m$ and Y is the m-dimensional space with elements $y = [y_1, \ldots, y_m]^T$.

Standard computation of solutions of the set of difference equation (11) shows that solution and information operator are linear and are given by

$$S(\lambda) = [A^{m-1}B, A^{m-2}B, \ldots, AB, B]\lambda \tag{16}$$

$$F(\lambda) = \begin{pmatrix} C & 0 & \cdots & 0 & 0 \\ CA & CB & \cdots & 0 & 0 \\ CA^2 & CAB & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ CA^{m-2} & CA^{m-3}B & \cdots & CB & 0 \\ CA^{m-1} & CA^{m-2}B & \cdots & CAB & CB \end{pmatrix} \lambda. \tag{17}$$

Example 3. Parameter estimation of multiexponential models. Let us consider the multiexponential model

$$y(t) = \sum_{i=1}^{l} \mu_i e^{-\nu_i t} + \rho(t) \tag{18}$$

where μ_i and ν_i are unknown real parameters and $\rho(t)$ is unknown but bounded by a given $\epsilon(t)$

$$|\rho(t) \leq \epsilon(t). \tag{19}$$

Suppose that m values $[y(t_i), \ldots, y(t_m)]$ are known and the aim is to estimate parameters μ_i and ν_i, $i = 1, \ldots, l$.

By setting $\xi_i = e^{-\nu_i}$, $i = 1, \ldots, l$, the space Λ is taken as the $2l$-dimensional space with elements

$$\lambda = [\mu_1, \ldots, \mu_l, \xi_1, \ldots, \xi_l]^T. \tag{20}$$

$S(\lambda)$ can be taken as the identity operator. In this way, estimation of variables ξ_is is considered instead of ν_is. Original variables can be obtained by logarithmic transformation.

Y is defined as the m-dimensional space with elements $y = [y(t_1), \ldots, y(t_m)]^T$. Then, information operator $F(\lambda)$ results to be the polynomial function

$$\begin{bmatrix} F_1(\lambda) \\ \cdots \\ F_m(\lambda) \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{l} \mu_i \xi_i^{t_1} \\ \cdots \\ \sum_{i=1}^{l} \mu_i \xi_i^{t_m} \end{bmatrix}. \tag{21}$$

Example 4. Multistep prediction with ARMA models. Consider the ARMA model (7) and suppose that the aim is to estimate y_{m+h} when past values $[y_1, \ldots, y_m]$ are measured (h-step ahead prediction problem). For the sake of notation simplicity, consider the case $h = 2$.

The space Λ can be defined as the $p + q + 2$-dimensional space with elements

$$\lambda = [v_1, \ldots v_p, \theta_1, \ldots, \theta_q, \rho_{m+1}, \rho_{m+2}]^T. \quad (22)$$

If no *a priori* knowledge on parameter λ is available, K is given by

$$K = \{\lambda \in \Lambda; |\rho_{m+1}| \leq \epsilon_{m+1}, |\rho_{m+2}| \leq \epsilon_{m+2}\}. \quad (23)$$

z is the 1-dimensional space with elements $z = y_{m+2}$ and consequently $S(\lambda)$ is the polynomial function given by

$$\begin{aligned} S(\lambda) = {} & (v_1 v_1 + v_2) y_m + (v_1 v_2 + v_3) y_{m-1} + \cdots \\ & + v_p y_{m-p+1} + \theta_1 u_{m+1} \\ & + (v_1 \theta_1 + \theta_2) u_m + \cdots + \theta_q u_{m-q+1} \\ & + v_1 \rho_{m+1} + \rho_{m+2}. \end{aligned} \quad (24)$$

Y is an $(m - p)$ dimensional space with elements $y = [y_{p+1}, \ldots, y_m]^T$ and $F(\lambda)$ is linear and given by

$$F(\lambda) = \begin{pmatrix} y_p & \cdots & y_1 & u_p & \cdots & u_{p+1-q} & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ y_{m-1} & \cdots & y_{m-p} & u_{m-1} & \cdots & u_{m-q} & 0 & 0 \end{pmatrix} \lambda. \quad (25)$$

MAIN DEFINITIONS AND CONCEPTS

This section provides basic definitions of the main sets involved in the theory, optimality concepts used to evaluate estimator's performances, types of estimators investigated.

Relevant sets

Now we introduce the following sets which play key roles in set membership estimation theory.

- MUS_y: Measurement Uncertainty Set

$$MUS_y = \{\tilde{y} \in Y : \|\tilde{y} - y\|_\infty^w \leq \epsilon\} \quad (26)$$

- EUS_ϕ: Estimate Uncertainty Set (for a given estimator ϕ)

$$EUS_\phi = \phi(MUS_y) \quad (27)$$

- FPS_y: Feasible Problem elements Set

$$FPS_y = \{\lambda \in K : \|y - F(\lambda)\|_\infty^w \leq \epsilon\} \quad (28)$$

- FSS_y: Feasible Solutions Set

$$FSS_y = S(FPS_y). \quad (29)$$

Note the difference between EUS_ϕ and FSS_y. The former depends on the particular estimator ϕ used and gives all possible estimated values that could be obtained for all possible measurements consistent with the actual measurement y and the given error bounds. The latter depends only on the problem setting and gives all possible values which are consistent with the available information on the problem.

In the literature on parameter estimation, where problem element λ represents the parameters to be estimated and $S(\lambda)$ is identity (see example 1), FPS_y coincides with FSS_y and has been given also different names such as feasible parameters set, membership-set estimate and likelihood set.

An exact description of FSS_y or EUS_ϕ is in general not simple, since they may be very complex sets (e.g. nonconvex, not simply connected). For this reason, approximate descriptions are often looked for, using simply shaped sets like boxes or ellipsoids containing (outer bounding) or contained in (inner bounding) the set of interest (see Fig. 2). In particular minimum volume outer box (MOB) or ellipsoid (MOE) and maximum volume inner box (MIB) or ellipsoid (MIE) are of interest. Information of great practical interest is also provided by the values uncertainty intervals (VUI) and estimate uncertainty intervals (EUI), giving the maximum ranges of possible variations of the feasible and

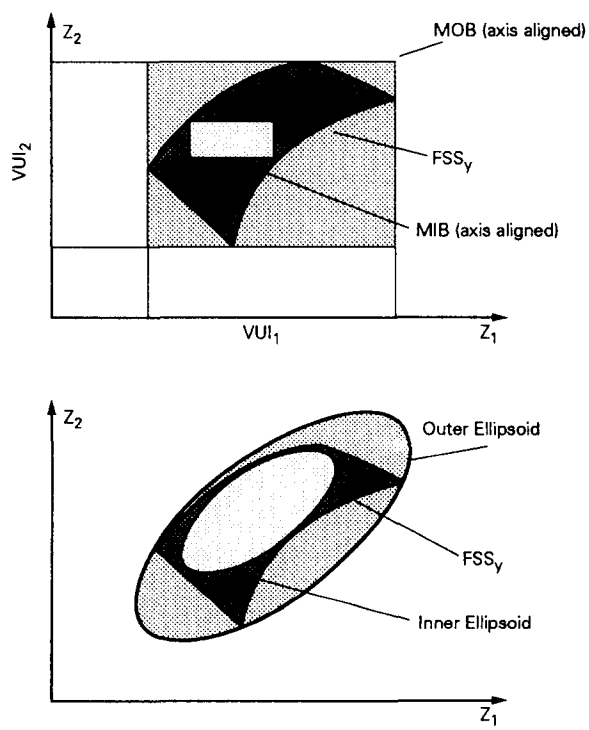

FIG. 2. Box and ellipsoid inner and outer bounding. FSS_y = Feasible Solutions Set, MOB = Maximal volume Outer Box, MIB = Minimal volume Inner Box and VUI_i = Uncertainty Interval of Variable z_i.

the estimated values, respectively. The VUIs are defined as

$$\mathrm{VUI}_i = [z_i^m, z_i^M] \quad i = 1, \ldots, l \tag{30}$$

where

$$z_i^m = \inf_{z \in FSS_y} z_i = \inf_{\lambda \in FPS_y} S_i(\lambda) \quad i = 1, \ldots, l$$
$$z_i^M = \sup_{z \in FSS_y} z_i = \sup_{\lambda \in FPS_y} S_i(\lambda) \quad i = 1, \ldots, l. \tag{31}$$

Note that the VUIs are the sizes (along coordinate axis) of the axis aligned box of minimal volume containing FSS_y (see Fig. 2).

The EUIs are defined in the same way substituting EUS_ϕ for FSS_y.

Errors and optimality concepts

Algorithm performance will be measured according to the following errors:

- Y-local error $E_y^\epsilon(\phi)$

$$E_y^\epsilon(\phi) = \sup_{\lambda \in FPS_y} \|S(\lambda) - \phi(y)\| \tag{32}$$

- Λ-local error $E_\lambda^\epsilon(\phi)$

$$E_\lambda^\epsilon(\phi) = \sup_{y \in MUS_{F(\lambda)}} \|S(\lambda) - \phi(y)\| \tag{33}$$

- global error $E^\epsilon(\phi)$

$$E^\epsilon(\phi) = \sup_{y \in Y} E_y^\epsilon(\phi) = \sup_{\lambda \in \Lambda} E_\lambda^\epsilon(\phi). \tag{34}$$

Dependence on ϵ will be dropped out in subsequent notation, except when necessary.

Algorithms minimising these types of errors will be indicated respectively as Y-locally, Λ-locally and globally optimal.

We notice that Y-local optimality is of particular interest in system identification problems, where a set of measurements y is available and one wants to determine the best achievable estimate $S(\lambda)$ for each possible y using an algorithm $\phi(y)$. Also Λ-local optimality is a particularly meaningful property in estimation problems, since it ensures the minimum uncertainty of the estimates for the worst measurement y, for any possible element $\lambda \in K$.

It is worthwhile to observe that Y- and Λ-local optimality are stronger properties than global optimality, as it can be seen from (34). In fact, for example, a Y-locally optimal algorithm minimises the local error $E_y(\phi)$ for all data y, while a globally optimal algorithm minimises the global error $E(\phi)$ only for the worst data. In other words, a Y-locally optimal algorithm is also globally optimal, while the converse is not necessarily true.

Classes of estimators

Now we introduce some classes of estimators whose properties have been investigated in the literature.

The first class is related to the idea of taking the Chebicheff center of FSS_y as estimate of $S(\lambda)$. The center of FSS_y, $c(FSS_y) \in Z$, and the corresponding radius, $\mathrm{rad}(FSS_y)$, are defined by

$$\sup_{z \in FSS_y} \|c(FSS_y) - z\| = \inf_{\tilde{z} \in Z} \sup_{z \in FSS_y} \|\tilde{z} - z\| = \mathrm{rad}\,(FSS_y). \tag{35}$$

A central estimator ϕ^c is defined as

$$\phi^c(y) = c(FSS_y). \tag{36}$$

The second class includes estimators analogous to unbiased estimators in statistical theory, which give exact values if applied to exact data.

An estimator ϕ is correct if

$$\phi(F(\lambda)) = S(\lambda) \quad \forall \lambda \in \Lambda. \tag{37}$$

Such class is meaningful only for $l \leq m$, that is when there are at least as many measurements as variables to be estimated (the typical situation in estimation practice). This class contains most of the commonly used estimators, such as for example projection estimators.

A projection estimator ϕ^p is defined as

$$\phi^p(y) = S(\lambda_y) \tag{38}$$

where $\lambda_y \in \Lambda$ is such that

$$\|y - S(\lambda_y)\| = \inf_{\lambda \in \Lambda} \|y - S(\lambda)\|. \tag{39}$$

The most widely investigated and used estimators in this class are least square estimators (ϕ^{LS}), which are projection estimators when an l_2 norm in used in space Y. Least absolute values and least maximum values estimators have been also considered in the literature, which are projection estimators when l_1 and l_∞ norms are respectively used in space Y.

In the next sections we will review the results available in the literature regarding the following aspects: existence and characterisation of estimators, optimal with respect to some of the optimality concepts introduced above; actual computation of the derived optimal estimators; evaluation of the errors of optimal and of projection estimators; and exact or approximate description of feasible sets FPS_y, FSS_y and estimate uncertainty set EUS_ϕ. Whenever possible, a statistical counterpart of the presented results will be indicated, based on the analogy:

Y-local optimality
$\Leftrightarrow$ minimum variance optimality

$FSS_y \Leftrightarrow$ minimum variance estimate pdf

$EUS_\phi \Leftrightarrow$ estimate pdf

EUIs $\Leftrightarrow$ estimate confidence intervals

VUIs $\Leftrightarrow$ Cramer–Rao lower bound confidence intervals.

NONLINEAR PROBLEMS

A first important result is related to the existence of a Y-locally optimal estimator. It is worth noting that no general results are available for Λ-locally optimal estimators. This result also shows that the minimum Y-local error is given by the radius of FSS_y.

Result 1 (Micchelli and Rivlin, 1977; Traub and Wozniakowski, 1980). A central estimator ϕ^c is Y-locally optimal

$$E_y(\phi^c) \leq E_y(\phi) \quad \forall y \in Y, \forall \phi. \tag{40}$$

Its Y-local error is

$$E_y(\phi^c) = \text{rad}\,(FSS_y). \tag{41}$$

□

This result holds for any norm in Λ, Z, Y.

It can be considered as the counterpart of the conditional mean theorem in statistics. As it happens for conditional mean estimators, central estimators are in general difficult to be actually computed. In fact, the computation of ϕ^c involves the knowledge of FSS_y, which may be a very complex set (nonconvex, not simply connected).

Several approaches have been proposed to describe FSS_y, mainly in papers related to dynamic system parameter estimation. In Keesman and van Straten (1988), a random sample of parameters is generated by a Monte Carlo technique, and (4) and (5) are used to check if they belong to FSS_y.

In Smit (1983) and Walter and Piet-Lahanier (1986), global optimisation methods based on random search are used to construct the boundary of FSS_y. In Smit (1983), projections of FSS_y onto coordinate one-dimensional or two-dimensional subspaces are looked for. In Walter and Piet-Lahanier (1986), intersections of the boundary of FSS_y with bundles of straight lines centred at points inside FSS_y are searched. The optimization methods used in these papers converge in probability to the global maximum or minimum of interest. However, this convergence property is not very useful in practice, because no estimate is given of the distance of the achieved solution from the global solution. Moreover, all these approaches suffer the curse of dimensionality. These reasons motivate interest in looking for less detailed but more easily computable information on FSS_y.

An important result in this direction is that the computation of ϕ^c and of its Y-local error do not require the exact knowledge of FSS_y, but only of the VUIs.

Result 2 (Milanese and Tempo, 1985). The center $c(FSS_y)$ can be computed as

$$c_i(FSS_y) = (z_i^M + z_i^m)/2 \quad i = 1, \ldots, l. \tag{42}$$

The radius rad (FSS_y) can be computed as

$$\text{rad}\,(FSS_y) = \max_i\,(z_i^M - z_i^m)/2 \tag{43}$$

where z_i^m and z_i^M are given by (31). □

Result 2 states that the computation of a central algorithm and of minimum Y-local error is equivalent to the computation of the VUIs, requiring the solution of only $2l$ optimisation problems of the type (31).

Problems (31) are in general not convex, exhibiting local extrema. Any of the general global optimization algorithms available in the literature give approximate solutions converging to the exact ones only in probability and, more seriously, they do not provide any assessment on how far the approximate solution is from the correct one.

If $S(\lambda)$ and $F(\lambda)$ are polynomial functions, specific global algorithms exist to obtain better results.

Result 3 (Milanese and Vicino, 1991). If $S(\lambda)$ and $F(\lambda)$ are polynomial, algorithms exist converging with certainty to global extrema of problems (31). □

Under the assumptions of Result 3, problems (31) are polynomial optimisation problems, in the sense that both cost functions and constraints are polynomials in λ. Polynomial problems are in general nonconvex and may admit local extrema (see e.g. Ecker, 1980). Nevertheless, if all the variables are strictly positive (in which case the term *signomial problems* is used), an algorithm is available to find a global maximum (Falk, 1973; Milanese and Vicino, 1988, 1991). The underlying idea of this algorithm is to construct iteratively a sequence of convex problems approximating better and better the original problem. In this way, the algorithm generates a sequence of lower and upper bounds of the global extremum, converging monotonically to it. If the sign of some of the variables is

not known, it is possible to reduce a polynomial problem to a signomial problem by setting these variables as the difference of strictly positive new variables.

The hypothesis of Result 3 covers large classes of problems, as shown by examples 1–4. Then Results 2 and 3 imply that an optimal estimator and its error can be exactly computed for several nonlinear problems of practical interest. No analogous result is available in the statistical context.

It must be remarked that most of the papers in the literature focus on studying FSS_y, while very few results are available on EUS_ϕ. We now report a result showing that, for any correct estimator, FSS_y is an inner bounding set of EUS_ϕ.

Result 4 (Milanese and Belforte, 1982). If ϕ is correct then

$$FSS_y \subseteq EUS_\phi \quad \forall y \in Y. \tag{44}$$

□

Hence, it results that for correct estimators the VUIs are lower bounds of the EUIs, that is

$$VUI_i \subseteq EUI_i \quad i = 1, \ldots, l. \tag{45}$$

Now we consider the properties of projection estimators. In general they are not optimal with respect to any of the three considered type of errors (Kacewicz *et al.*, 1986). However they are "almost" Y-locally optimal (within a factor 2) as shown by the following result.

Result 5 (Kacewicz *et al.*, 1986). A projection algorithm ϕ^P is such that

$$E_y(\phi^P) \leq 2 \operatorname{rad}(FSS_y) \leq 2E_y(\phi) \quad \forall y \in Y, \forall \phi. \tag{46}$$

□

Projection estimators enjoy interesting properties of robustness with respect to inexact knowledge of the uncertainty bound ϵ. Central estimators are not robust in such a sense: a central algorithm computed supposing that $\epsilon = \epsilon_0$ may not be optimal if the actual ϵ is different. A central estimate $\phi^c(y)$ may not even belong to the actual FSS_y and its Y-local error $E_y(\phi^c)$ may be greater than $2 \operatorname{rad}(FSS_y)$.

On the contrary, projection estimators preserve their almost Y-locally optimality properties in presence of bounded uncertainty on ϵ as shown by the next result.

Result 6 (Tempo and Wasilkowski, 1988). Let ϕ^P be the projection estimator computed with $\epsilon = \epsilon_0$. Then

$$E_y^\epsilon(\phi^P) \leq 2 \operatorname{rad}(FSS_y) \leq 2E_y^\epsilon(\phi) \quad \forall y \in Y, \forall \phi, \forall \epsilon \leq \epsilon_0. \tag{47}$$

□

Result 6 can be rephased saying that projection estimators are robustly almost Y-locally optimal.

Note that projection estimators have nice properties also in a statistical context. For example, an l_2-projection estimator is the Maximum Likelihood Estimator (MLE) if noise ρ is supposed Gaussian, an l_1-projection estimator is the MLE if noise is supposed to have Laplace pdf, an l_∞-projection estimator is a MLE if noise is uniformly distributed. Also l_1 and l_∞-projection estimators have interesting robustness properties with respect to uncertainty in the pdf's knowledge (Launer and Wilkinson, 1979; Poljak and Tsypkin, 1980; van den Bos, 1988).

LINEAR PROBLEMS

In this section we consider the case in which $S(\lambda)$ and $F(\lambda)$ are linear. In this case, equation (4) is written as

$$y = \mathscr{A}\lambda + \rho \tag{48}$$

where $\mathscr{A}$ is a matrix of dimension (m, n).

These assumptions are restrictive, but include cases of practical interest such as parameter estimation of linear regressions, parameter estimation of ARMA models with polynomial trends and harmonic components, state estimation of dynamic systems, time series forecasting. Moreover, if uncertainty bounds are not too large, linear theory can be used for a first approximate analysis using some linearization techniques.

From Result 1 we know that in general a central estimator is Y-locally and globally optimal. Next result shows that in the linear case it is also correct and Λ-locally optimal in the class of correct estimators.

Result 7 (Milanese *et al.*, 1986).

- ϕ^c is correct
- ϕ^c is Y-locally optimal

$$E_y(\phi^c) \leq E_y(\phi) \quad \forall y, \forall \phi \tag{49}$$

- ϕ^c is Λ-locally optimal (among correct estimators)

$$E_\lambda(\phi^c) \leq E_\lambda(\phi) \quad \forall \lambda \in K, \forall \phi \text{ correct}. \tag{50}$$

□

In Kacewicz *et al.* (1986) it is proven that Result 7 holds for any norm in Y.

Note that under the present assumptions, FSS_y and FPS_y are polytopes. Then from Result 2 it follows that ϕ^c and its Y-local error $E_y(\phi^c)$ can be obtained by solving the $2l$ linear programming problems (31).

The next result shows that a linear estimator can be computed which is correct, globally optimal and Λ-locally optimal within the class of correct estimators. This result gives a complete solution to the linear case, representing the counterpart of the Gauss–Markov theory in statistical estimation.

Result 8 (Milanese and Belforte, 1982; Milanese and Tempo, 1985). Let $K = \Lambda$ and $m \geqslant n$. Then there exists a linear estimator H^* such that

- H^* is correct
- H^* is globally optimal

$$E(H^*) \leq E(\phi) \quad \forall \phi \tag{51}$$

- H^* is Λ-locally optimal (among correct estimators)

$$E_\lambda(H^*) \leq E_\lambda(\phi) \quad \forall \lambda \in \Lambda,\ \forall \phi \text{ correct} \tag{52}$$

- Its errors are

$$\begin{aligned} E(H^*) &= E_\lambda(H^*) = E_{\mathscr{A}\lambda}(H^*) \\ &= \operatorname{rad}(FSS_{\mathscr{A}\lambda}) \quad \forall \lambda \in \Lambda. \end{aligned} \tag{53}$$

□

Estimator H^* can be computed from the knowledge of the active constraints of the linear programming problems (31) with $y = 0$ (Milanese and Belforte, 1982; Milanese and Tempo, 1985).

In case that an l_2 norm is used in Y, H^* can be computed by least squares. In fact, under this assumption, a least squares estimator is linear and correct, robustly Y-locally optimal and Λ-locally optimal within the class of correct estimators, as shown by the following result.

Result 9 (Kacewicz *et al.*, 1986). Let $K = \Lambda$, $m \geq n$ and Y be a Hilbert space. Let ϕ^{LS} be the projection (Least Square) estimator computed with $\epsilon = \epsilon_0$. Then

- ϕ^{LS} is central
- ϕ^{LS} is linear and correct
- ϕ^{LS} is robustly Y-locally optimal

$$E_y^\epsilon(\phi^{LS}) \leq E_y^\epsilon(\phi) \quad \forall y \in Y,\ \forall \phi,\ \forall \epsilon \leq \epsilon_0 \tag{54}$$

- ϕ^{LS} is Λ-locally optimal (among correct estimators)

$$E_\lambda(\phi^{LS}) \leq E_\lambda(\phi) \quad \forall \lambda \in \Lambda,\ \forall \phi \text{ correct.} \tag{55}$$

□

The sets FPS_y, FSS_y and EUS_ϕ, (for linear ϕ), are polytopes described by the sets of linear inequalities appearing in their definitions (28), (29) and (27). This is not the simplest way to describe them (for example, many linear inequalities may not concur to defining the boundary of the polytope) and simpler descriptions could be of interest. One way of characterising a polytope P is through its vertices. Algorithms exist which allow one to compute recursively the vertices of a polytope P_k, defined by the first k measurements, from the knowledge of P_{k-1} and the kth measurement (Kapitonenko, 1982; Piet-Lahanier and Walter, 1988; Mo and Norton, 1988; Piet-Lahanier and Walter, 1989). It is interesting to note that the number of vertices may be much smaller than $2l^m$ (theoretical maximum). For example, Monte Carlo simulations on ARMA models parameter estimation (Mo and Norton, 1988) have shown that the mean number of vertices of FSS_y is approximately constant for $m > 50$. For $l = 4$ and $l = 5$, for example, they are approximately 5 and 15, respectively.

Polytopes can be represented alternatively by describing their faces. In Broman and Shensa (1988) this representation is used to derive a recursive algorithm. This approach seems more involved than the previous one, but it allows also the recursive computation of an outer bounding polytope with a fixed number of faces, leading to an approximating description of the polytope of interest by means of a polytope of prescribed complexity.

The most investigated approaches to approximate description of polytopes are through ellipsoid and box bounds for the case of parameter estimation, where the polytope of interest is the feasible parameter set.

A recursive algorithm for outer bounding ellipsoid computation has been proposed in Fogel and Huang (1982). The underlying idea is as follows.

Let OE_{k-1} be the outer ellipsoid bounding P_{k-1}. Let R_k be the feasible parameter set corresponding only to the kth measurement

$$R_k = \{\lambda \in \Lambda : y_k - \epsilon_k \leq a_k^T \lambda \leq y_k + \epsilon_k\} \tag{56}$$

where a_k^T is the kth row of $\mathscr{A}$.

Clearly $P_k \subseteq OE_{k-1} \cap R_k$. OE_k is computed as the minimal volume ellipsoid containing $OE_{k-1} \cap R_k$, and then containing P_k also.

If an ellipsoid OE_k is defined by its centers λ_k^c and positive definite matrix Σ_k according to

$$OE_k = \{\lambda \in \Lambda : (\lambda - \lambda_k^c)^T \Sigma_k^{-1} (\lambda - \lambda_k^c) \leq 1\} \tag{57}$$

the following recursive algorithm has been obtained

Result 10 (Fogel and Huang, 1982). OE_k can be computed by the recursion

$$\lambda_k^c = \lambda_{k-1}^c + \frac{\sigma_k V_k a_k v_k}{\epsilon_k^2} \tag{58}$$

$$\Sigma_k = \left(1 + \sigma_k - \frac{\sigma_k v_k^2}{\epsilon_k^2 + \sigma_k \mu_k}\right) V_k \tag{59}$$

where

$$V_k = \Sigma_{k-1} - \frac{\sigma_k \Sigma_{k-1} a_k a_k^T \Sigma_{k-1}}{\epsilon_k^2 + \sigma_k \mu_k} \tag{60}$$

$$v_k = y_k - a_k^T \lambda_{k-1} \tag{61}$$

$$\mu_k = a_k^T \Sigma_{k-1} a_k \tag{62}$$

and σ_k is the positive solution of the equation

$$(l-1)\mu_k^2 \sigma_k^2 + [(2l-1)\epsilon_k^2 - \mu_k + v_k^2]\mu_k \sigma_k + \epsilon_k^2[l(\epsilon_k^2 - v_k^2) - \mu_k] = 0 \tag{63}$$

if a positive solution exists, otherwise $\sigma_k = 0$.

□

Considerations on computational complexity of this algorithm and on slight modifications for implementation on a systolic architecture can be found in Deller (1989). A modification of this algorithm with data-dependent updating and forgetting factor has been proposed in Dasgupta and Huang (1987).

A similar approach can be used for the recursive computation of inner bounding ellipsoids (Norton, 1987, 1989b). Let IE_{k-1} the inner bounding ellipsoid contained in P_{k-1}. Then IE_k is chosen as the maximal volume ellipsoid such that

$$IE_k \subseteq IE_{k-1} \cap R_k \subseteq P_k. \tag{64}$$

The resulting recursive algorithm is much the same as for the outer bounding ellipsoid and is not reported here.

The main drawback of these recursive algorithms is that they do not give the minimal and maximal volume ellipsoids bounding the feasible parameter set (Belforte and Bona, 1985; Norton, 1987). This is true also for improved versions of the algorithm (Belforte and Bona, 1985; Pronzato *et al.*, 1989). Since IE_k has an unfortunate tendency to shrink rapidly and vanish (Norton, 1989b), it turns out that the inclusion $IE_k \subset P_k \subset OE_k$ in practice may not give any reasonable information on how loose bound OE_k is.

A nonrecursive solution to the problem of finding the Minimal volume Outer Ellipsoid contained in FPS_y (MOE_{FPS}) and the Maximal volume Inner Ellipsoid contained in FPS_y (MIE_{FPS}), has been proposed in (Vicino and Milanese (1989, 1991). The solution for MIE_{FPS} is given by the following result.

Result 11 (Vicino and Milanese, 1989). The MIE_{FPS} has center λ^{c*} and matrix Σ^* solution of

$$\max \det(\Sigma) \tag{65}$$

subject to

$$\lambda^c, \Sigma: \begin{cases} (u_i^T \lambda^c + c_i)^2 - u_i^T \Sigma u_i \geq 0, & i = 1, \ldots, 2m \\ u_i^T \lambda^c + c_i \geq 0, & i = 1, \ldots, 2m \\ \Sigma_i > 0, & i = 1, \ldots, n \end{cases}$$

where Σ_i, $i = 1, \ldots, n$ are the principal minors of Σ, and matrix $U \in \mathbf{R}^{(2m,n)}$ (with rows denoted by u_i^T) and vector $c \in \mathbf{R}^{2m}$ are given by

$$U = [A^T \mid -A^T]^T; \quad c = [-y_-^T \mid y_+^T]^T \tag{66}$$

$$\begin{aligned} y_- &= [y_1 - \epsilon w_1, y_2 - \epsilon w_2, \ldots, y_m - \epsilon w_m]^T \\ y_+ &= [y_1 + \epsilon w_1, y_2 + \epsilon w_2, \ldots, y_m + \epsilon w_m]^T. \end{aligned} \tag{67}$$

□

Problem (65) is clearly a polynomial optimisation problem and can be solved by use of signomial programming. It must be remarked that the solution of (65) may be computationally cumbersome, even for few parameters. Then less general but simpler solutions may be of interest. For example, the maximum ellipsoid of given shape may be looked for. This corresponds to considering that Σ is given except for a scale factor (for example the shape of the outer ellipsoid given by Result 10 can be used). In such case, problem (65) reduces to a linear programming problem with $(n+1)$ variables and $(2m+1)$ constraints.

The solution for MOE_{FPS} also can be obtained by solving a suitable polynomial problem (Vicino and Milanese, 1991). Unfortunately, the computational complexity is high for the general case, and does not reduce, as for MIE_{FPS}, if restricted classes of ellipsoids are considered.

Finally we report results on the computation of extremal volume inner and outer boxes.

Here boxes are defined as follows

$$B(\lambda^c, l, R) = \{\lambda \in \Lambda : \|R(\lambda - \lambda^c)\|_\infty^l \leq 1\} \tag{68}$$

where R is an orthonormal matrix. In this way the box is described by its center λ^c, axis lengths l_i and rotation R. If $R = I$ the box is aligned with coordinate axis.

A solution to the problem of finding the Minimal volume Outer box contained in FPS_y (MOB_{FPS}) is provided in Vicino and Milanese (1991), as the solution of a suitable polynomial problem. Its computational complexity is high, unless the rotation of the box is given. In such case the problem can be reduced to a linear

programming problem. In particular, if $R = I$ the axis aligned MOB_{FSS} can be computed directly from (31), requiring the solution of $2l$ linear programming problems with n variables and $2m$ inequalities constraints.

The solution to the problem of finding the Maximal volume Inner Box contained in FPS_y (MIB_{FPS}) is provided by the following result.

Result 12 (*Vicino and Milanese,* 1989). The MIB_{FPS} has center λ^{c^*}, axis length l^* and rotation R^* solution of

$$\max \prod_{i=1}^{n} l_i \tag{69}$$

subject to

λ^c, l, R:

$$\begin{cases} l_i > 0, \ i = 1, \ldots, n \\ (u_i^T R^T \lambda^c + c_i) - \sum_{j=1}^{n} l_j \, |u_{ij}| \geq 0, \ i = 1, \ldots, 2m. \\ R^T R = I \end{cases}$$

□

Problem (69) is clearly a polynomial optimisation problem which can be solved by use of signomial programming. If matrix rotation R is fixed, problem (69) reduces to a convex problem with $2n$ variables and $(2m + n)$ linear constraints, which can be efficiently solved by means of normally available convex programming algorithms. If axis length l is also fixed except for a scale factor (i.e. the maximum box of given shape is looked for), problem (69) reduces to a linear programming problem with $(n + 1)$ variables and $(2m + 1)$ constraints.

OTHER TYPES OF RESULTS

In this section we briefly recall papers on topics related to set membership estimation theory, such as experiment design, estimation with reduced order models, uncertainty in the information operator. Almost all these papers consider linear problems.

Experiment design

In the previous sections information operator $\mathscr{A}$ is supposed given. In some practical application it is possible to choose among different information operators $\mathscr{A}$ (optimal information problem). For example it may be possible to choose the sampling times at which measurements are taken of the input and the output of the dynamic system to be identified. Then a natural choice is the one minimising the error $E_\lambda(\phi^c)$. In Belforte *et al.*, (1987) some results are given for the case in which information is provided by sampling. In Milanese *et al.* (1986) similar results are derived for more general classes of information. In the latter paper it is also shown that the optimal sampling times can be chosen *a priori,* and no improvements can be obtained by means of more sophisticated sampling schemes. In Micchelli (1989) the optimal sampling problem is approached through p-widths theory.

Another criterion is to minimise the volume of FPS_y. In Norton (1987), a recursive selection procedure is given, based on heuristics which avoids poor choices without guaranteeing the best. In Pronzato and Walter (1989), a characterization is given of the minimum number of sampling times assuring minimum volume of the feasible parameter set FPS_y for $y = \mathscr{A}\lambda$.

Reduced order models

In the previous sections, it is supposed that the structure of the problem is given, for example the number of autoregressive and moving average terms for an ARMA model. In many cases, however, the structure of the problem and in particular the dimension of space Λ is not known and must be evaluated from the available information (order determination problems). In Norton (1989a) and Veres and Norton (1989), some methods are discussed. Some of them are analogous to methods widely used for order determination in statistical contexts, such as the principal component analysis and singular value decomposition. A method is also proposed, based on the expected behaviour of FPS_y for overparameterised and underparameterised structures.

A second important problem is how estimation algorithms can take into account that only approximating structures are used. The usual approach in statistical contexts is to ignore the deterministic nature of modelling errors, and eventually discard badly approximating structures with residuals evidently not satisfying the assumed statistical hypotheses. In the UBB approach, modelling errors can be taken into account in a more natural way, since it is possible to evaluate bounds on such modelling errors (Genesio and Milanese, 1976; Lau *et al.*, 1990; Krause *et al.*, 1990). A deeper analysis is carried out in Kacewicz *et al.* (1988), where it is considered explicitly that using approximating structures corresponds to restrict the analysis to a subset $K \subset \Lambda$ not containing the "true" problem element λ. The concept of conditionally central estimator is introduced as an extension of central estimator, and it is shown that it is Y-locally optimal. In the same paper it is also

shown that there are two possible ways of extending least squares estimators. The first one corresponds to what is usually done (more or less explicitly) when dealing with reduced order models. However, this estimator does not preserve any of the interesting optimality properties of least squares estimators. A second type of extension is introduced, which is shown to have interesting Λ-locally and Y-locally optimality properties.

Uncertain information operator

In some papers the case is studied, in which information matrix $\mathscr{A}$ is not exactly known. In particular perturbation of the type $\mathscr{A} = \mathscr{A}_0 + \Delta\mathscr{A}$ have been considered, where $\mathscr{A}_0$ is given and $\Delta\mathscr{A}$ is not known but bounded. In Clement and Gentil (1988), a modification of the recursive algorithm for outer ellipsoid bounding reported in Result 10 is proposed. In Tempo *et al.* (1988), and in Belforte *et al.* (1988), two different extensions of FPS_y are considered. In the former reference FPS_y^1 is defined by considering that (28) holds for all $\Delta\mathscr{A}$ and is described by a set of $m2^{n+1}$ linear inequalities. In the latter, FPS_y^2 is defined by considering that (28) holds for some $\Delta\mathscr{A}$, and the problem of finding the corresponding MOB by means of suitable linear programming problems is also discussed.

APPLICATIONS

The UBB approach is now beginning to be used in various application fields. Without the aim of giving an exhaustive list of references, we cite some papers reporting applications to real word problems arising in biology (Belforte *et al.*, 1983; Norton, 1986), pharmacokinetics (Milanese and Belforte, 1982; Gomeni *et al.*, 1986), time series prediction (Vicino *et al.*, 1987), economy (Milanese *et al.*, 1988), chemistry (Walter *et al.*, 1986), image processing (Venot *et al.*, 1986), ecology (Van Straten, 1985; Keesman and Van Straten, 1987), measurement (Smit, 1983; Belforte *et al.*, 1987, 1988; Verhoof and Smit, 1988), tracking (Broman and Shensa, 1986), speech processing (Deller and Luk, 1987; Deller and Hsu, 1987; Combettes and Trussel, 1988; Deller, 1989).

Application of set membership estimation theory have also been investigated in the context of identification for robust and adaptive control design (Dasgupta and Huang, 1987; Kosut, 1988; Vicino *et al.*, 1990; Lau *et al.*, 1990; Krause *et al.*, 1990; Fiorio *et al.*, 1990; Ljung and Wahlberg, 1990).

CONCLUSIONS

In this paper an outline of the main results in the area of estimation theory for set membership uncertainty has been presented. The main emphasis of the paper is on the following aspects: existence and characterisation of worst-case optimal estimators; actual computation of the derived optimal estimators; evaluation of the errors of optimal and of other widely used estimators (least squares, least absolute values, least maximum values); exact or approximate description of feasible parameter and solution sets. A quick reference to less assessed topics such as experiment design, reduced order modeling, more general error models is also made in the paper.

Some general considerations may be drawn from this overview. Concerning linear problems, it is apparent that real advances have been made in the last decade. As a result of this, it comes out that properties of estimators and exact or approximate description of feasible parameter and solution sets can be considered subjects with reasonably well understood and usable solutions. In fact, many of the available algorithms have been used for several applications in different real world problems.

Concerning nonlinear problems, in spite of the work done in the last few years, much more remains to be done. In fact, some algorithms for computing exact parameter or solution uncertainty intervals have been proposed and work reasonably well on problems with a limited number of measurements and parameters. However, their behavior in more complex situations has not been deeply investigated yet.

Several basic problems remain open and need a thorough investigation, both for linear and nonlinear problems, for example the topological properties of the feasible parameter set as a function of the nonlinearity and uncertainty structures, inner and outer bounding for the nonlinear case, the effects of model approximations, the interaction of set membership estimation theory and robust or adaptive control.

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